

MIDTERM ASSESSMENT

Answer the four problems. I: 30% of the total mark, II: 20% of the total mark, III and IV: 25% of the total mark each.

I – RICARDIAN EQUIVALENCE

Consider a two-period economy populated with a continuum of measure 1 of identical households. There is unique non storable good. Preferences of household i are

$$U = \log c_{i1} + \beta \log c_{i2}$$

with $\beta \in]0, 1[$.

Each household receives endowments $\omega_1 > 0$ in period 1 and $\omega_2 > 0$. They have access to bonds, denoted B_i with (gross) interest rate R .

There is a government who taxes endowments in the periods at rates $\tau_1 > 0$ and $\tau_2 > 0$ and distribute a check $\theta > 0$ in period 1. Taxes and checks are the same for all then agents. The government can also issue debt B .

- 1 – Write and solve for the optimal problem of a representative household
- 2 – Write the government period 1, period 2 and intertemporal budget constraints.
- 3 – Define a competitive equilibrium
- 4 – Solve for competitive equilibrium price and quantities
- 5 – Assume that the government increases the check distributed in period 1 and finances it by increasing τ_2 while leaving τ_1 constant. How does this affect price, bonds holdings and consumption quantities? Explain.

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Let us now assume that a mass $\delta \in]0, 1[$ of agents dies at the end of period 1, and that they are informed of their death at the beginning of period 1 before taking any decision. These agents are denoted the *unlucky agents* “ u ”. They are replaced in period 2 by a mass δ of *newborn* (“ b ”). Finally, a mass $1 - \delta$ of agents live for the 2 periods (the *lucky ones* “ ℓ ”). All the agents alive in a given period receive the same endowments, the same check if in period 1 and pay the same taxes. Utilities are $\log c_{\ell 1} + \beta \log c_{\ell 2}$ for the lucky agents, $\log c_{u1}$ for the unlucky ones and $\beta \log c_{b2}$ for the newborn.

- 6 – Solve for the optimal behaviour of the three types of agents.
- 7 – Define a competitive equilibrium
- 8 – Assume that the government increases the check distributed in period 1 and finance it by increasing τ_2 while leaving τ_1 constant. How does this affect price and quantities? Explain.

II – TWO AGENTS WITH COMPLETE MARKETS AND REDUNDANT ASSET

An economy consists of two infinitely lived consumers named $i = 1, 2$. There is one nonstorable consumption good. Consumer i consumes c_t^i at time t . Consumer i ranks consumption streams by

$$\sum_{t=0}^{\infty} \beta^t u(c_t^i),$$

where $\beta \in (0, 1)$ and $u(c)$ is increasing, strictly concave, and twice continuously differentiable. Consumer 1 is endowed with a stream of the consumption good

$$y_t^1 = 1, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, \dots$$

Consumer 2 is endowed with with a stream of the consumption

$$y_t^2 = 0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, \dots$$

Assume that there are complete markets with time-0 trading.

- 1 – Define a competitive equilibrium.
- 2 – Compute a competitive equilibrium.
- 3 – Suppose that one of the consumers markets a derivative asset that promises to pay $0 < d < 1$ units of consumption in periods $0, 2, 4, \dots$ and $-d$ in periods $1, 3, 5, \dots$. What would the price of that asset be?

III – STOCHASTIC GROWTH WITH ELASTIC LABOR SUPPLY

Suppose the planner seeks to maximize

$$E \left\{ \sum_{t=0}^{\infty} \beta^t u(c_t, l_t) \right\}, \quad 0 < \beta < 1,$$

subject to the resource constraint

$$c_t + k_{t+1} = z_t f(k_t, l_t) + (1 - \delta)k_t, \quad 0 < \delta < 1,$$

with initial conditions $k_0 > 0$ and $z_0 > 0$. Productivity z_t evolves according to a Markov process $F(z'|z)$.

The planner chooses (c_t, l_t) . Utility is increasing and concave in c , decreasing and convex in l . Production is increasing, concave, and CRS.

- 1 – Let $v(k, z)$ denote the planner's value function. Set up the Bellman equation.
- 2 – Discuss briefly how the contraction mapping theory helps us solve this problem. In particular, is the solution v unique? What condition, if any, would guarantee that the optimal plan is unique?
- 3 – Derive the planner's optimality conditions for c , k' , and l .
- 4 – Now assume no uncertainty and

$$u(c, l) = \log c - \frac{l^{1+\phi}}{1+\phi}, \quad f(k, l) = zk^\alpha l^{1-\alpha}.$$

Compute the deterministic steady state. Explain the effect of a permanent increase in non-stochastic productivity z on capital-to-output ratio, labour, and consumption-to-output ratio, providing some economic intuition for your findings.

5 – Provide a sketch of pseudo code for value function iteration to solve this problem. How is uncertainty reflected in the problem of this kind? Is the presence of endogenous labour choice an issue? How can you deal with that?

6 – Now write the same problem (with no uncertainty) in continuous time: specifically, write the HJB equation and derive the optimality conditions (i.e. the optimal labour condition and the Euler Equation).

7 – It is tedious but straightforward to show that the $\dot{c} = 0$ schedule takes the form

$$c = k^{-\phi} \left[\left(\frac{\alpha}{\delta + \rho} \right)^{\frac{1-\alpha}{\phi+\alpha}} (1 - \alpha) z^{\frac{1+\phi}{\phi+\alpha}} \right].$$

Taking this result as given (so no need to derive it), draw the phase diagram, assuming the $\dot{k} = 0$ equation is given by the usual $c = zk^\alpha - \delta k$ (i.e. assume that ϕ is large and so labour supply is almost inelastic, so its effect on the $\dot{k} = 0$ can be ignored (otherwise it's a bit messy!)).

Trace out the dynamics following a permanent shock to productivity z .

IV – TWO COUNTRIES IN THE AIYAGARI MODEL

Consider the stationary equilibrium of the Aiyagari model covered in class. The representative firm's production function is $Y = ZK^\alpha N^{1-\alpha}$ with $\alpha \in (0, 1)$ and where K and N are the aggregate quantities of capital and labor and Z is aggregate productivity. Consider two countries, A and B , which are identical in all respects except for their inhabitant's discount rates, in particular $\rho_A > \rho_B$. Denoting the stationary equilibrium interest rates by r_A^* and r_B^* and the stationary capital stocks by K_A^* and K_B^* how do you expect these to differ across the two countries when the countries are in autarky? What will happen to their interest rates, capital stocks, and net foreign asset positions when these countries integrate financially? Draw a diagram to illustrate.

END OF EXAM